

Principles of Data Communications and Networks Modeling (CS4865/6865)

Lab Report #1: Introduction to network simulation I

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Exercise 2:

2.1 Code

```
1  set ns [new Simulator]
2
3  set nf [open exer_2.nam w]
4  $ns namtrace-all $nf
5
6  proc finish {} {
7      global ns nf
8      $ns flush-trace
9      close $nf
10     exec nam exer_2.nam &
11     exit 0
12 }
13
14 set n0 [$ns node]
15 set n1 [$ns node]
16
17 $ns duplex-link $n0 $n1 1Mb 10ms DropTail
18
19 set udp0 [new Agent/UDP]
20 $ns attach-agent $n0 $udp0
21
22 set cbr0 [new Application/Traffic/CBR]
23 $cbr0 set packetSize_ 500
24 $cbr0 set interval_ 0.005
25 $cbr0 attach-agent $udp0
26
27 set null0 [new Agent/Null]
28 $ns attach-agent $n1 $null0
29
30 $ns connect $udp0 $null0
```

```

31
32 $ns at 0.5 "$cbr0_start"
33 $ns at 4.5 "$cbr0_stop"
34
35 $ns at 5.0 "finish"
36
37 $ns run

```

2.2 NAM Screenshot

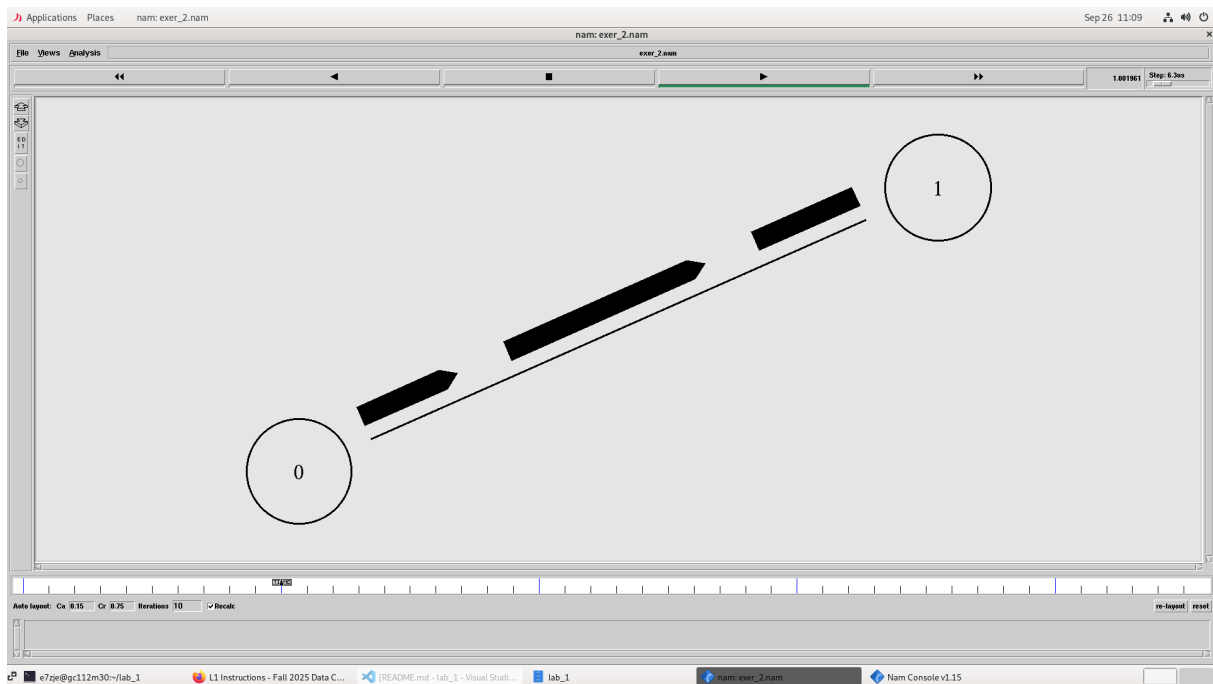


Figure 1: NAM Screenshot of Exercise 2

Exercise 3:

3.1 Trace file

```

1 V -t * -v 1.0a5 -a 0
2 A -t * -n 1 -p 0 -o 0x7fffffff -c 30 -a 1
3 A -t * -h 1 -m 1073741823 -s 0
4 n -t * -a 0 -s 0 -S UP -v circle -c black -i black
5 n -t * -a 1 -s 1 -S UP -v circle -c black -i black
6 l -t * -s 0 -d 1 -S UP -r 1000000 -D 0.01 -c black
7 + -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0
   0 ----- null}
8 - -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0
   0 ----- null}

```

```

9  h -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0
    -1 ----- null}
10 + -t 0.505 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    1.0 1 ----- null}
11 - -t 0.505 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    1.0 1 ----- null}
12 h -t 0.505 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    1.0 -1 ----- null}
13 + -t 0.51 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    1.0 2 ----- null}
14 - -t 0.51 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    1.0 2 ----- null}
15 h -t 0.51 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    1.0 -1 ----- null}
16 r -t 0.514 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0
    1.0 0 ----- null}
17 + -t 0.515 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    1.0 3 ----- null}
18 - -t 0.515 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    1.0 3 ----- null}
19 h -t 0.515 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    1.0 -1 ----- null}
20 r -t 0.519 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    1.0 1 ----- null}
21 + -t 0.52 -s 0 -d 1 -p cbr -e 500 -c 0 -i 4 -a 0 -x {0.0
    1.0 4 ----- null}
22 - -t 0.52 -s 0 -d 1 -p cbr -e 500 -c 0 -i 4 -a 0 -x {0.0
    1.0 4 ----- null}
23 h -t 0.52 -s 0 -d 1 -p cbr -e 500 -c 0 -i 4 -a 0 -x {0.0
    1.0 -1 ----- null}
24 r -t 0.524 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    1.0 2 ----- null}
25 + -t 0.525 -s 0 -d 1 -p cbr -e 500 -c 0 -i 5 -a 0 -x {0.0
    1.0 5 ----- null}
26 - -t 0.525 -s 0 -d 1 -p cbr -e 500 -c 0 -i 5 -a 0 -x {0.0
    1.0 5 ----- null}
27 h -t 0.525 -s 0 -d 1 -p cbr -e 500 -c 0 -i 5 -a 0 -x {0.0
    1.0 -1 ----- null}
28 r -t 0.529 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    1.0 3 ----- null}
29 + -t 0.53 -s 0 -d 1 -p cbr -e 500 -c 0 -i 6 -a 0 -x {0.0
    1.0 6 ----- null}
30 - -t 0.53 -s 0 -d 1 -p cbr -e 500 -c 0 -i 6 -a 0 -x {0.0
    1.0 6 ----- null}

```

3.2 First Packet Transmitted

The first packet being transmitted can be seen in line number 7.

```
+ -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0 0 ----- null}
```

3.3 First Packet Received

The first packet being transmitted can be seen in line number 16.

```
r -t 0.514 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 1.0 0 ----- null}
```

Exercise 4:

4.1 Code

```
1  set ns [new Simulator]
2
3  set nf [open exer_4.nam w]
4  $ns namtrace-all $nf
5
6  set n0 [$ns node]
7  set n1 [$ns node]
8  set n2 [$ns node]
9
10 $ns duplex-link $n0 $n1 1Mb 10ms DropTail
11 $ns duplex-link $n1 $n2 1Mb 10ms DropTail
12
13 set udp0 [new Agent/UDP]
14 $ns attach-agent $n0 $udp0
15
16 set cbr0 [new Application/Traffic/CBR]
17 $cbr0 set packetSize_ 500
18 $cbr0 set interval_ 0.005
19 $cbr0 attach-agent $udp0
20
21
22 set null0 [new Agent/Null]
23 $ns attach-agent $n2 $null0
24
25 $ns connect $udp0 $null0
26
27 $ns at 0.5 "$cbr0 start"
28 $ns at 4.5 "$cbr0 stop"
29
30 proc finish {} {
31     global ns nf
32     $ns flush-trace
33     close $nf
34     exec nam exer_4.nam &
```

```

35     exit 0
36 }
37
38 $ns at 5.0 "finish"
39
40 $ns run

```

4.2 NAM Screenshot

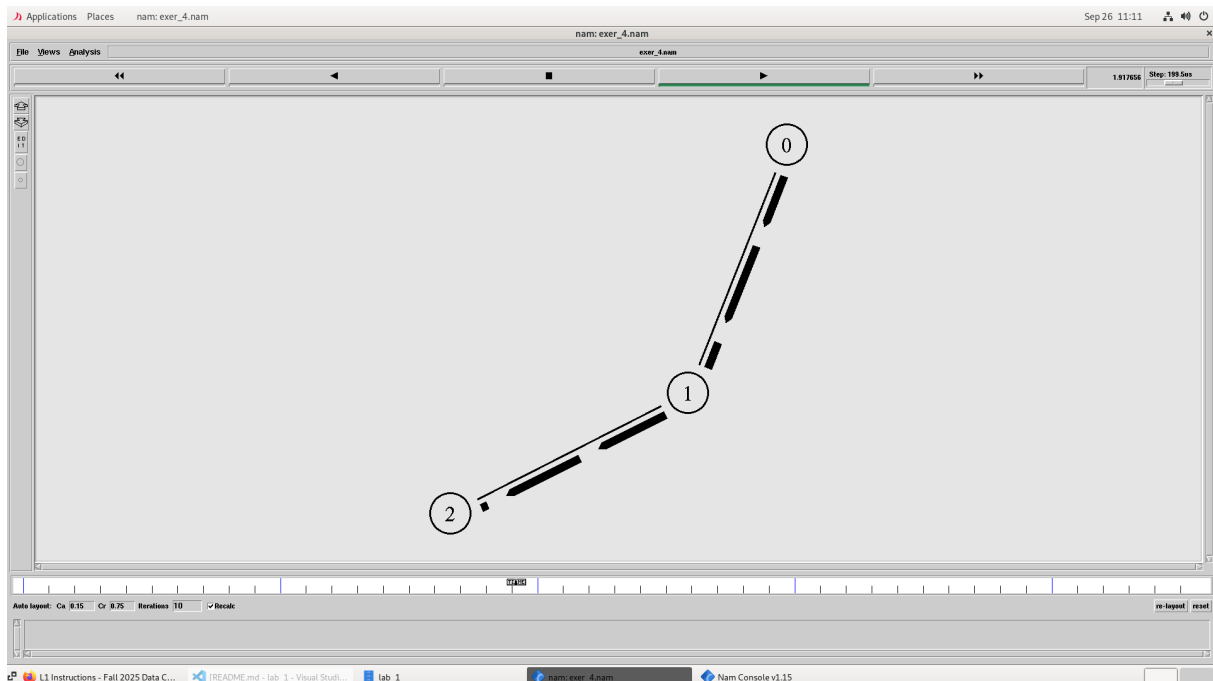


Figure 2: NAM Screenshot of Exercise 4

Exercise 5:

5.1 Trace file

```

1  V -t * -v 1.0a5 -a 0
2  A -t * -n 1 -p 0 -o 0x7fffffff -c 30 -a 1
3  A -t * -h 1 -m 1073741823 -s 0
4  n -t * -a 0 -s 0 -S UP -v circle -c black -i black
5  n -t * -a 1 -s 1 -S UP -v circle -c black -i black
6  n -t * -a 2 -s 2 -S UP -v circle -c black -i black
7  l -t * -s 0 -d 1 -S UP -r 1000000 -D 0.01 -c black
8  l -t * -s 1 -d 2 -S UP -r 1000000 -D 0.01 -c black
9  + -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 2.0
    0 ----- null}
10 - -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 2.0
    0 ----- null}

```

```

11 h -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 2.0
    -1 ----- null}
12 + -t 0.505 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 1 ----- null}
13 - -t 0.505 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 1 ----- null}
14 h -t 0.505 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 -1 ----- null}
15 + -t 0.51 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    2.0 2 ----- null}
16 - -t 0.51 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    2.0 2 ----- null}
17 h -t 0.51 -s 0 -d 1 -p cbr -e 500 -c 0 -i 2 -a 0 -x {0.0
    2.0 -1 ----- null}
18 r -t 0.514 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0
    2.0 0 ----- null}
19 + -t 0.514 -s 1 -d 2 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0
    2.0 0 ----- null}
20 - -t 0.514 -s 1 -d 2 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0
    2.0 0 ----- null}
21 h -t 0.514 -s 1 -d 2 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0
    2.0 -1 ----- null}
22 + -t 0.515 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    2.0 3 ----- null}
23 - -t 0.515 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    2.0 3 ----- null}
24 h -t 0.515 -s 0 -d 1 -p cbr -e 500 -c 0 -i 3 -a 0 -x {0.0
    2.0 -1 ----- null}
25 r -t 0.519 -s 0 -d 1 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 1 ----- null}
26 + -t 0.519 -s 1 -d 2 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 1 ----- null}
27 - -t 0.519 -s 1 -d 2 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 1 ----- null}
28 h -t 0.519 -s 1 -d 2 -p cbr -e 500 -c 0 -i 1 -a 0 -x {0.0
    2.0 -1 ----- null}
29 + -t 0.52 -s 0 -d 1 -p cbr -e 500 -c 0 -i 4 -a 0 -x {0.0
    2.0 4 ----- null}
30 - -t 0.52 -s 0 -d 1 -p cbr -e 500 -c 0 -i 4 -a 0 -x {0.0
    2.0 4 ----- null}

```

5.2 First Packet Transmitted

The first packet being transmitted can be seen in line number 9 and the time stamp is 0.5

```
+ -t 0.5 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 2.0 0 ----- null}
```

5.3 First Packet Received at Forwarding Node

The first packet being received at forwarding node can be seen in line number 18 and the time stamp is 0.514

```
r -t 0.514 -s 0 -d 1 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 2.0 0 ----- null}
```

5.4 First Packet Received at Destination Node

The first packet being received at destination node can be seen in line number 39 and the time stamp is 0.528

```
r -t 0.528 -s 1 -d 2 -p cbr -e 500 -c 0 -i 0 -a 0 -x {0.0 2.0 0 ----- null}
```

5.5 Delivery Time Calculation

From trace data:

- Destination receive time = 0.528 seconds
- Source transmission time = 0.5 seconds (enqueue time represents when transmission begins)

The total delivery time of a packet from the source node to the destination node can be calculated using the following formula:

$$\begin{aligned}\text{Delivery Time} &= \text{Destination Receive Time} - \text{Source Transmission Time} \\ &= 0.528 - 0.5 \\ &= 0.028 \text{ seconds}\end{aligned}$$

This delivery time is derived from the timestamps in the trace file, which show the complete path of the packet:

1. Packet 0 is enqueued at node 0 at 0.5 seconds.
2. Packet 0 is received at node 1 (the forwarding node) at 0.514 seconds.
3. Packet 0 is received at node 2 (the destination node) at 0.528 seconds.

The total time includes the transmission delays from node 0 to node 1 and from node 1 to node 2, along with any processing delays at the intermediate node.

This analysis represents a simple three-node topology, where node 0 is the source, node 1 is the intermediate forwarding node, and node 2 is the destination. The traffic is CBR (Constant Bit Rate), and the packet flows from node 0 to node 2 via node 1.

Exercise 6: Theoretical Delay Calculation

6.1 Network Parameters from the Trace File

From the trace, we can extract:

```
Link 0 -> 1: 1 -t * -s 0 -d 1 -S UP -r 1000000 -D 0.01 -c black
Link 1 -> 2: 1 -t * -s 1 -d 2 -S UP -r 1000000 -D 0.01 -c black
Packet size: 500 bytes (from -e 500 in packet events)
```

Key parameters:

```
Link bandwidth (rate): 1,000,000 bps (1 Mbps)
Link propagation delay: 0.01 seconds
Packet size: 500 bytes = 4,000 bits
```

6.2 Delay Components Formula

Total delay = Transmission delay + Propagation delay + Processing delay + Queuing delay

For our case:

- Processing delay is typically negligible in NS-2 unless specified.
- Queuing delay should be zero for the first packet (no packets ahead in queue)

$$\text{Transmission delay} = \frac{\text{Packet size}}{\text{Bandwidth}}$$

$$\text{Propagation delay} = \frac{\text{Distance}}{\text{Propagation speed}}$$

6.3 Theoretical Calculation

For one hop (node 0 → node 1):

$$\begin{aligned}\text{Transmission delay} &= \frac{4000 \text{ bits}}{1,000,000 \text{ bps}} \\ &= 0.004 \text{ seconds}\end{aligned}$$

Propagation delay = 0.01 seconds

Total per hop delay = 0.004 + 0.01 = 0.014 seconds

For two hops (node 0 → node 1 → node 2):

Hop 1 (0 → 1): 0.014 seconds

Hop 2 (1 → 2): 0.014 seconds

Total theoretical delay = 0.014 + 0.014 = 0.028 seconds

6.4 Comparison with Empirical Measurement

Empirical delay measured = 0.528s - 0.5s = 0.028 seconds

Theoretical delay (0.028s) = Empirical delay (0.028s)

Yes, this calculated delay is same as the delay value which was calculated in Exercise 5.2

References

- [1] Mark Greis, Tutorial for the Network Simulator "ns": <http://www.isi.edu/nsnam/ns/tutorial/index.html>
- [2] TCL description: <http://en.wikipedia.org/wiki/Tcl>
- [3] ns manual: <https://www.isi.edu/nsnam/ns/ns-documentation.html>
- [4] Trace file format: <https://www.isi.edu/websites/nsnam/ns/doc/node289.html>
- [5] nam trace file format: <https://www.isi.edu/websites/nsnam/ns/doc/node616.html>